Count-specific quartic majorants on a continuous parameter cell

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Abstract

We give a finite exact certificate criterion for a conditional Berry–Esseen comparison throughout a probability and threshold cell. A separate quartic for each binomial count uses one common pair of conditional noise laws. Signed second-moment information restricts their variances to a polygon. A finite-sample fourth-moment bound makes the upper expectation convex on that polygon. This note proves the reduction; acquisition of its hypotheses and completion of a numerical cover are separate obligations.

1 The conditional problem

Throughout, Φ and ϕ are the standard normal distribution function and density, and $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$. Write $Y = B + U$, where $\mathbb{P}(B=1) = p$, $q = 1-p$, and $\mathbb{E}(U \mid B = i) = 0$. The two conditional laws U_0, U_1 are fixed throughout every calculation. Let $n \geq 2$ be an integer and assume $A_0, A_1 \geq 0$. Put

$$s_i = \mathbb{E}U_i^2, \quad x = qs_0, \quad y = ps_1, \quad v = pq, \quad w = v + x + y, \quad \lambda = n(x+y).$$

Then $X = (Y-p)/\sqrt{w}$ is centered with variance one. Assume

$$-A_0 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq A_1, \qquad 0 \leq a_i \leq r_i, \quad a_0 \leq p, \quad a_1 \leq q,$$

$$0 \leq \lambda \leq L, \qquad \rho = \mathbb{E}|X|^3 \geq \rho_- > 0, \qquad b_- \leq \beta := \mathbb{E}[X|X|] \leq b_+.$$

The sign restrictions ensure $X \leq 0$ in class zero and $X \geq 0$ in class one. On a closed probability cell $[l,h] \subset (0,1/2)$ choose support ceilings A_i, r_i independently of p. For instance, $a_0 + a_1 \leq d$ permits $r_0 = \min(d,h)$, $r_1 = \min(d, 1-l)$; dropping the remaining joint restriction enlarges the domain.

For a phase $u \in [u_-, u_+]$ and integer k, the target is the positive closed-event score

$$\frac{\sqrt{n}}{\rho}\left\{\mathbb{P}\left(\sum_{i=1}^{n} Y_i \leq k+u\right) - \Phi\left(\frac{k+u-np}{\sqrt{nv+\lambda}}\right)\right\} < C_{\mathrm{E}}. \tag{1}$$

This is one oriented score. Both signs of the original discrepancy require their own acquisition or the justified reflection argument.

2 One common variance polygon

The sign partition gives the exact identity

$$\beta = \frac{v(1-2p) - x + y}{v + x + y}. \tag{2}$$

Indeed the positive and negative conditional square contributions are $p(q^2+s_1)$ and $q(p^2+s_0)$. Since the denominator is positive, the beta bounds are equivalent to

$$\begin{aligned} -(1+b_+)x+(1-b_+)y &\le v[b_+-(1-2p)],\\ (1+b_-)x-(1-b_-)y &\le v[(1-2p)-b_-]. \end{aligned}$$

Replace each right side by a rational upper bound for its cubic on $[l,h]$. Together with $x,y\ge 0$ and $x+y\le L/n$, these halfplanes define a fixed compact outer polygon $\mathcal{V}$. Every actual common variance pair belongs to it, though not every point need be realizable. Intersecting each pair of boundary lines and retaining the intersections satisfying every halfplane gives all vertices. A reader may require a nonempty two-dimensional polygon and reject degenerate cases. Empty-domain exclusion needs its own exact proof.

3 Finite common-moment tests

For $j=0,\dots,n$ choose quartics $P_j(z)=\sum_{r=0}^4 c_{jr}z^r$, with $c_{j4}\ge 0$, satisfying

$$P_j(z)\ge 1 \quad (z\le 0), \qquad P_j(z)\ge 0 \quad (z\ge 0). \tag{3}$$

Let W_j sum $n-j$ independent U_0 variables and j independent U_1 variables. Conditional independence gives

$$\mathbb{P}\left(\sum Y_i\le k+u\right)\le \sum_{j=0}^n \pi_j(p)\mathbb{E}P_j(W_j+j-k-u), \qquad \pi_j(p)=\binom{n}{j}p^jq^{n-j}. \tag{4}$$

The left condition in (3) applies at zero, retaining every event-plane atom. All expectations use the same two conditional laws.

Lemma 1 (Support and variance reduction)**.** *For each fixed probability and phase, the right side of (4) is bounded by the maximum of the following evaluations: a vertex (x,y) of $\mathcal{V}$ and one pair*

$$\gamma_0\in\{-A_0,0,r_0\}, \qquad \gamma_1\in\{A_1,0,-r_1\}. \tag{5}$$

For that evaluation define

$$\begin{aligned} t_j &= (n-j)x/q+jy/p,\\ M_{j0}&=1, \quad M_{j1}=0, \quad M_{j2}=t_j,\\ M_{j3}&=(n-j)\gamma_0x/q+j\gamma_1y/p,\\ M_{j4}&=(n-j)\gamma_0^2x/q+j\gamma_1^2y/p+3(1-1/n)t_j^2,\\ E_j&=\sum_{r=0}^4 c_{jr}\sum_{i=0}^r\binom{r}{i}(j-k-u)^{r-i}M_{ji}, \qquad F=\sum_{j=0}^n\pi_j(p)E_j. \end{aligned}$$

Proof. Put $\mu_i=\mathbb{E}U_i^3$. Integrating the nonnegative support polynomials $(U_0+A_0)(a_0-U_0)U_0^2$ and its reflected counterpart gives

$$\mathbb{E}U_0^4\le(a_0-A_0)\mu_0+A_0a_0s_0, \qquad \mathbb{E}U_1^4\le(A_1-a_1)\mu_1+A_1a_1s_1. \tag{6}$$

Independence and zero means give

$$\mathbb{E}W_j^4=(n-j)\mathbb{E}U_0^4+j\mathbb{E}U_1^4+3t_j^2-3\{(n-j)s_0^2+js_1^2\}.$$

Cauchy's inequality for the n individual variances bounds the quantity in braces below by t_j^2/n. All leading coefficients and binomial weights are nonnegative, so this gives an upper bound in (4).

For $s_0 > 0$, $\gamma_0 = \mu_0/s_0 \in [-A_0, a_0]$. The mixture bound is affine in γ_0. At its endpoint $-A_0$, the coefficient of s_0 in (6) is A_0^2, independent of a_0. At the other endpoint it is a_0^2. The remaining expression is convex quadratic in a_0, hence is bounded by its values at $0, r_0$. The reflected argument treats class one. This proves the nine common choices (5). A zero variance makes that conditional variable vanish and requires no division.

For each choice (5), F is affine in x, y except for a sum of nonnegative multiples of squares t_j^2. Thus it is convex on $\mathcal{V}$, and its value at a convex combination of vertices is no larger than the largest vertex value. The displayed fourth moment is an extremal evaluation of an upper expression, not the false claim $\mathbb{E}U_i^4 \le \gamma_i^2 s_i$ for an arbitrary actual law. □

4 Continuous cells and exact acceptance

Let T_- be a rational lower bound, uniform on the cell, for

$$\Phi\left(\frac{k+u-np}{\sqrt{npq+\lambda}}\right) + \frac{C_{\mathrm{E}}\rho_-}{\sqrt{n}}. \tag{7}$$

The Gaussian part must retain the numerator's sign. Put $a = k + u_- - nh$, $v_- = l(1-l)$, $v_+ = h(1-h)$. A valid lower normal argument is $a/\sqrt{nv_+ + L}$ for $a \ge 0$, and $a/\sqrt{nv_-}$ for $a < 0$. Outward square-root and Gaussian bounds then give (7). For a negative numerator, using only the upper noise endpoint would give the wrong inequality.

Proposition 2 (Exact cell criterion)**.** *Suppose (3) holds globally and, for every polygon vertex and every pair (5), the rational polynomial*

$$R(p,u) = p^2(1-p)^2\{T_- - F(p,u)\} \tag{8}$$

is strictly positive on $[l,h] \times [u_-, u_+]$. Then (1) holds on the entire cell for every actual conditional pair satisfying the stated hypotheses.

Proof. The cleared denominator is positive. The lemma and (4) give $\mathbb{P}(\sum Y_i \le k+u) < T_-$. Subtract the Gaussian CDF and apply (7), followed by $\rho \ge \rho_-$. At zero noise the same expressions are the literal polynomial expectations at the actual threshold; no jump is discarded and no variance is divided out. □

The coordinate degrees in (8) are at most $n+4$ in p and four in u. The moments have denominators at most p^2q^2; the largest degree is from $p^2q^2\pi_j(p)$ times a constant moment. For $n = 5$ the bounds are $(9, 4)$. After mapping the cell to $[0,1]^2$, a polynomial $\sum r_{ij} s^i t^j$ has tensor Bernstein coefficients

$$b_{ab} = \sum_{i \le a,\, j \le b} r_{ij} \frac{\binom{a}{i}}{\binom{9}{i}} \frac{\binom{b}{j}}{\binom{4}{j}}.$$

Positive coefficients imply (8), since the basis is nonnegative and sums to one. Inconclusive coefficients do not disprove positivity. For (3), set $z = \pm t/(1-t)$ and multiply the appropriate ray polynomial by $(1-t)^4$. Exact positive endpoints and zero Sturm roots on $[0,1]$ prove positivity on the ray. Repeated-factor cases may conservatively reject; unresolved roots cannot accept. Floating-point optimization only proposes coefficients. Exact acceptance reconstructs these polynomials and the complete cover.

4.1 Keeping the Gaussian term at the same noise level

The uniform target in (7) can lose useful information by taking its minimum at a different noise level from the largest quartic expectation. For a negative normal numerator there is a direct remedy.

Proposition 3 (Concave Gaussian correction)**.** *Suppose throughout a cell*

$$a_{\max} = k + u_+ - nl \le 0, \qquad a_{\min}^2 = (k + u_- - nh)^2 \le 3nl(1-l). \tag{9}$$

Then the cell criterion remains valid if its target is chosen separately for each polygon vertex (x, y), using a lower bound for

$$\Phi\left(\frac{k+u-np}{\sqrt{npq + n(x+y)}}\right) + \frac{C_{\mathrm{E}}\rho_-}{\sqrt{n}}. \tag{10}$$

A rational lower Gaussian value can use numerator $a_{\min}$ and variance $nl(1-l) + n(x+y)$. It is constant on the probability/phase cell for this fixed vertex, so the same polynomial degree bound applies.

Proof. For fixed p, u set $a = k + u - np$, $D = npq + \lambda$, and $g(\lambda) = \Phi(a/\sqrt{D})$. Direct differentiation gives

$$g'(\lambda) = -\frac{a\phi(a/\sqrt{D})}{2D^{3/2}}, \qquad g''(\lambda) = \frac{a\phi(a/\sqrt{D})}{4D^{5/2}}\left(3 - \frac{a^2}{D}\right).$$

Condition (9) implies $a \le 0$ and $a^2 \le 3npq$, so g is concave on the entire noise interval, including the constant case $a = 0$. For each common skew choice, $F - g(n(x+y))$ is therefore convex on the polygon. Its maximum occurs at a vertex. Check (10) at each vertex and apply the same argument as before. For a negative numerator, increasing either the numerator or the denominator increases its standardized value. This justifies the stated uniform lower bound. No endpoint choice for the normal term is borrowed from a different variance pair. □

4.2 Keeping the same law's third absolute moment

The lower bound $\rho \ge \rho_-$ may also discard useful information. The sign partition gives an exact expression for the third absolute moment before normalization:

$$Q := \mathbb{E}|Y - p|^3 = v(1-2v) + 3(px + qy) - \gamma_0 x + \gamma_1 y, \qquad \rho = \frac{Q}{(v+x+y)^{3/2}}. \tag{11}$$

Here $\gamma_i s_i = \mathbb{E}U_i^3$; a zero variance contributes zero. Indeed, expand $q\mathbb{E}(p - U_0)^3 + p\mathbb{E}(q + U_1)^3$ and use the zero conditional means. The actual Q is nonnegative.

Fix a noise slab $[L_-, L_+]$ and a rational upper bound $W \ge v + x + y$ on it. Let $c_0 > 0$ and $c_1 > 0$ be rational lower bounds for $C_{\mathrm{E}}/\sqrt{n}$ and $c_0/W^{3/2}$, respectively. For any fixed rational $\theta \in [0, 1]$,

$$\frac{C_{\mathrm{E}}\rho}{\sqrt{n}} \ge (1-\theta)c_0\rho_- + \theta c_1 Q. \tag{12}$$

This follows by taking a convex combination of $\rho \ge \rho_-$ and $\rho \ge Q/W^{3/2}$. Directed square-root bounds suffice to choose c_0, c_1. The sign of Q justifies rounding c_1 down.

The cell test can therefore use the right side of (12) in place of the constant third-moment contribution in (7) or (10). To justify the same finite reductions, first subtract $\theta c_1 Q$ from the upper expectation on actual laws. For fixed variances this expression is affine in each

γ_i. At an inward skew endpoint the new contribution is affine in a_i, leaving the nonnegative quadratic coefficient from the quartic majorant unchanged. Thus the same nine endpoint choices apply. For each choice, Q is affine in x, y, so subtracting it preserves the convexity used at the polygon vertices, including the Gaussian correction under (9). After multiplication by p^2q^2, the new term has degree at most eight in p; the bound $(9, 4)$ for $n = 5$ is unchanged.

An artificial endpoint evaluation of (11) need not be a moment of a probability law and can be negative. This causes no problem: nonnegativity was used only to derive (12) for actual laws. The resulting polynomial upper expression is then maximized on an enlarged domain. One must not round its coefficient again according to a sign inferred from a fictitious law.

4.3 Using acquired mean and sign information

Two further restrictions already present in the selected original contract tighten the same fixed polygon and endpoint bounds. Let $m = \mathbb{E}X_+ = v/\sqrt{w}$, where the positive part follows from the sign partition. If $m \geq m_- > 0$, then

$$\lambda \leq n\left(\frac{v^2}{m_-^2} - v\right). \tag{13}$$

If $2v_- > m_-^2$, the right side increases with v throughout the cell. We may replace the slab ceiling by

$$L_+^{\mathrm{eff}} = \min\left\{L_+,\, n\left(\frac{v_+^2}{m_-^2} - v_+\right)\right\}, \qquad W = v_+ + L_+^{\mathrm{eff}}/n. \tag{14}$$

Intersect the polygon also with $L_-/n \leq x + y \leq L_+^{\mathrm{eff}}/n$. This still covers every actual law in the original slab.

Suppose the original standardized conditional supports satisfy $X \leq -\varepsilon_0$ in class zero and $X \geq \varepsilon_1$ in class one, with $\varepsilon_i > 0$. Since $X = (B + U - p)/\sqrt{w}$, these conditions imply

$$a_0 \leq p - \varepsilon_0\sqrt{w}, \qquad a_1 \leq q - \varepsilon_1\sqrt{w}.$$

Retain also the acquired ceiling $a_0 + a_1 \leq d$. If s_- is a rational lower bound for $\sqrt{v_- + L_-/n}$, valid fixed inward ceilings are

$$r_0 = \min(d, h - \varepsilon_0 s_-), \qquad r_1 = \min(d, 1 - l - \varepsilon_1 s_-). \tag{15}$$

The reader checks their nonnegativity. It rejects empty or degenerate polygons rather than treating a failed calculation as an exclusion. These are consequences of acquired hypotheses; their use does not independently prove those hypotheses.

5 The selected row and the complete-cover obligation

The original five-summand BN probability-panel38 contract supplies

$$p \in [19/50, 39/100], \quad \rho \in [11/10, 23/20], \quad L = 7687/25000,$$

$$(b_-, b_+) = (-15293/500000, 307221/1000000).$$

$$A_0 = 82453/125000, \qquad A_1 = 703209/1000000.$$

These are acquired inputs, not consequences of the criterion. For (13)–(15) the same contract supplies

$$m_- = 217391/500000, \quad d = 471533/500000, \quad \varepsilon_0 = 29529/10^6, \quad \varepsilon_1 = 9459/200000.$$

The original raw threshold bound $|H| \le 2051019/10^6$ and $\sqrt{w} \le 547171/10^6$ imply, for the Bernoulli-coordinate threshold $h_B = 5p + \sqrt{w}H$,

$$\frac{777741882751}{10^{12}} \le h_B \le \frac{3072258117249}{10^{12}}.$$

Thus $1/2 < h_B < 7/2$, and only nearest thresholds $k = 1, 2, 3$ occur. Both closed phase halves must be retained for each. At half-integers both representations may be included. Mixture counts $j = 0, \dots, 5$ have a different role and do not justify this threshold restriction.

A completed row certificate must cover every mapped point or exclude it using a proved original necessary condition. A finite tree must retain both children at each split, all roots and all shared faces. Missing roots or children, duplicated root selections, and malformed objects reject. The earlier fixed-probability, fixed-phase pilot does not fulfill this obligation. The accompanying progress records identify which continuous regions have actually been accepted; an incomplete cover cannot justify removing a manuscript argument.

In this setting a certificate consists of a finite subdivision tree, six rational quartics and a rational weight θ at every leaf. Its mathematical claim is the score inequality (1) for every actual law in the specified acquired row. A floating-point linear program finds candidate coefficients. A separate standard-library Python reader recomputes the ray and tensor polynomials with exact rational arithmetic, checks strict inequalities, and checks both closed children at every split. A source-bound collector fixes the required threshold roots independently of the candidate data and runs the reader afresh on each. Thus a favorable search result or a stored success message is insufficient. The analytic reduction above, the original acquisition, and the exact readers are distinct parts of the proof available for human inspection.